

Advanced (Carolina Reaper)

Q1. Rationalize the denominator of $\frac{2}{\sqrt{3}}$.

- (A) $\frac{2\sqrt{3}}{3}$
- (B) $\frac{2}{3\sqrt{3}}$
- (C) $\frac{\sqrt{3}}{2}$
- (D) $\frac{2}{\sqrt{3}}$
- (E) $\frac{3\sqrt{3}}{2}$

Q2. Simplify $\frac{1}{\sqrt{5}-1}$ by rationalizing the denominator.

- (A) $\frac{1+\sqrt{5}}{4}$
- (B) $\frac{\sqrt{5}-1}{4}$
- (C) $\frac{\sqrt{5}+1}{4}$
- (D) $\frac{5-\sqrt{5}}{4}$
- (E) $\frac{5+\sqrt{5}}{4}$

Q3. A bookshelf is $5\sqrt{2}$ feet tall. If it is placed $2\sqrt{3}$ feet away from the wall, what is the distance from the top of the bookshelf to the point on the wall directly above it, after rationalizing the denominator?

- (A) $\sqrt{47}$
- (B) $\sqrt{43}$
- (C) $\sqrt{23}$
- (D) $\sqrt{37}$
- (E) $\sqrt{53}$

Q4. To rationalize the denominator of $\frac{\sqrt{7}}{\sqrt{5}-\sqrt{3}}$, what value should be multiplied to both numerator and denominator?

- (A) $\sqrt{5} + \sqrt{3}$
- (B) $\sqrt{7} + \sqrt{3}$
- (C) $\sqrt{5} - \sqrt{7}$
- (D) $\sqrt{7} - \sqrt{5}$
- (E) $\sqrt{3} - \sqrt{7}$

Q5. If $x = \frac{2}{\sqrt{3}+1}$ and $y = \frac{2}{\sqrt{3}-1}$, what is $x+y$ after rationalizing the denominators?

- (A) $\frac{4}{3}$
- (B) $\frac{4}{2}$
- (C) $\frac{2}{3}$
- (D) $\frac{8}{3}$
- (E) $\frac{2}{1}$

Q6. A bakery sells a cake for $15\sqrt{2}$ dollars. If a person buys two of these cakes and a drink for $2\sqrt{5}$ dollars, what is the total cost after rationalizing the denominator?

- (A) $30\sqrt{2} + 2\sqrt{5}$
- (B) $30 + 2\sqrt{5}$
- (C) $15\sqrt{2} + 2\sqrt{5}$
- (D) $32\sqrt{2}$
- (E) $30\sqrt{2} + \frac{2\sqrt{5}\cdot\sqrt{5}}{5}$

Q7. A jogger runs $8\sqrt{3}$ meters in $2\sqrt{5}$ seconds. What is his speed after rationalizing the denominator?

- (A) $\frac{4\sqrt{15}}{5}$
- (B) $\frac{4\sqrt{15}}{10}$
- (C) $4\sqrt{3}$
- (D) $\frac{4}{\sqrt{5}}$
- (E) $4\sqrt{15}$

Q8. To rationalize the denominator of $\frac{3}{\sqrt{2}-\sqrt{5}}$, which value should be multiplied to the numerator and denominator?

- (A) $\sqrt{2} + \sqrt{5}$
- (B) $\sqrt{5} - \sqrt{2}$
- (C) $\sqrt{3} + \sqrt{2}$
- (D) $\sqrt{3} - \sqrt{2}$
- (E) $\sqrt{5} + \sqrt{3}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator and simplify $\frac{\sqrt{11}+\sqrt{5}}{\sqrt{11}-\sqrt{5}}$. Provide the final simplified form.

Q10. Prove that $\frac{1}{\sqrt{a}-\sqrt{b}} = \frac{\sqrt{a}+\sqrt{b}}{a-b}$ by rationalizing the denominator. Explain the steps involved.

Chef's Tips

- Remind students to simplify their answers after rationalizing the denominators.
- Ensure students understand that rationalizing the denominator may involve multiplying the numerator and denominator by the conjugate of the denominator.
- Instruct students to check their answers for any remaining radicals in the denominator.